Firas Al Haddad

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Education

Concordia University Expected May 2027
B.Eng. Software Engineering (Co-op) | GPA: 3.47/4.3

Professional Experience

Data Analytics Intern – Data Engineering Focus May 2026 – Present
Pratt & Whitney Canada

- Built a modular Python ETL pipeline consolidating SharePoint and REST API data into reusable Parquet datasets for Power BI dashboards.
- Designed a five-layer medallion architecture (Landing to Platinum), with centralized orchestration, reusable API clients, and modular transformations.
- Integrated metal pricing data sources (Fastmarkets and Argus); standardized foreign currencies using S&P FX rates.
- Enhanced Power BI metal-pricing dashboards with historical comparisons, monthly/yearly averages, and period-over-period price changes.
- Established Azure DevOps branching and pull-request workflows; onboarded cross-regional teammates to Git-based collaboration.
- Extended Python ETL pipelines and Power BI dashboards to report CPI, PPI, and FX-adjusted economic indices for supplier-transition analysis.

Software Engineering Intern May 2025 – Aug 2025
Fonex Data Systems

- Built an internal Python/Tkinter desktop application to inspect, read, and reconfigure SFP optical transceivers through a hardware device.
- Implemented asynchronous Bluetooth Low Energy communication with `bleak/asyncio` for protected memory-page reads and writes.
- Designed a multithreaded connection manager to coordinate device I/O with the desktop interface; added transceiver photo capture.
- Tested alpha/beta mobile app releases with QMetry and performed optical-transceiver temperature checks in the hardware lab.

Data Analytics Intern Jan 2025 – Apr 2025
Pratt & Whitney Canada

- Consolidated 20+ Databricks data sources into a small set of curated datasets for spare-parts and manufacturing sales reporting.
- Refactored PySpark ingestion, cleaning, and transformations into reusable pipeline-compatible functions with automated tests.
- Redesigned Power BI views to surface daily, weekly, and monthly sales trends and progress toward business targets.

Projects

Campus Guide | *React Native, TypeScript, Google Maps, Firebase* GitHub

- Developed features for a Google Maps-based campus navigation app, integrating indoor room-to-room directions, cross-campus shuttle routing, travel-time estimates, and navigation tests.

IRIS (In Progress) | *React, TypeScript, Django, PostgreSQL, Docker* GitHub

- Co-developing a nonprofit web app for verifying low-income households and reserving surplus food and goods for pickup; built repeatable Django seed data and household-expiry tests.

Peer Review | *React, Java, Spring Boot, PostgreSQL* GitHub

- Developed role-aware student/instructor evaluation interfaces and REST API integrations; contributed sign-in unit tests and CI workflow improvements.

Focus Tracker | *React, TypeScript, TensorFlow.js* GitHub

- Developed React interfaces for a study-focus web app that monitors attention through webcam-based facial tracking, provides distraction alerts, and records session metrics.

Technical Skills

Languages: Python, SQL, Java, TypeScript/JavaScript, C#
Data Engineering: PySpark, Pandas, Databricks, ETL, Medallion Architecture, Parquet, Data Modeling
Development & BI: REST APIs, React, React Native, Spring Boot, PostgreSQL, Power BI, DAX, Power Query
Tools & Quality: Git, Azure DevOps, Docker, CI/CD, Unit Testing, Data Validation, SonarQube, QMetry
Other: English & French | Volleyball leagues, Custom PC Building, Strength Training, Strategy Games